HackSHSID 2026

Track 3 Problem Statement · An AI Assistant for Flight Disruptions at Key Airports

Problem setters: Accenture

> **Design an AI assistant for a fictional Chinese domestic airline that identifies problems, explains its reasoning and recommends responses when weather changes or operational disruptions occur at key airports, helping aviation professionals make decisions.**

1. The Problem

Changes in weather or disruptions to airport operations can affect subsequent flights, coordination with airports and passenger arrangements. Airlines need to identify problems promptly and, even when information is incomplete or contradictory, determine which impacts deserve attention, what to verify first and when to escalate the situation.

We want you to act as an external consulting team for this airline and design an AI assistant using public aviation data and simulated events. It should compare airports' normal operating levels, interpret new disruption scenarios, explain the basis for its assessments and gaps in information, recommend actions in order of priority, and produce separate draft messages for operations staff, airport coordination teams and passengers.

There is no standard answer. Which airports to focus on, how to define a disruption, how to prioritize checks and actions, and how to measure the assistant's value are all for you to decide and explain. AI provides decision support only: decisions about flight safety, flight release, cancellations, dispatch, maintenance and compensation must be made by qualified airline professionals. Draft passenger messages must also be reviewed by a person before they are published.

2. Evaluating Your Results

We care most about how you justify your approach and whether the assistant can handle new scenarios. Your submission should answer questions including but not limited to the following:

- How do you define a disruption? What evidence supports your definition when compared with the airport's own history, similar airports or an industry baseline?
- Can the assistant handle previously unseen events, rather than simply return answers written in advance?

- Can every key assessment be traced to data or rules? Can the assistant distinguish known facts, assumptions and unknown information?
- Do its recommendations clearly identify what to verify, the order of actions, escalation conditions and points requiring human decisions?
- For the same event, are messages to operations staff, airport coordination teams and passengers accurate and appropriate for each audience?
- When data conflicts, information is insufficient or a person rejects its recommendation, how does the assistant adapt and record feedback and improvements?
- What value does your solution offer compared with existing practices? How do you measure it, and under what circumstances would it fail?

How to present your results most convincingly is also up to you. Test the assistant on previously unseen scenarios, and use cases involving incorrect assessments, conflicting data or human rejection of recommendations to explain its limitations and improvements.

3. Reference Resources

Students may refer to the following public data sources, along with the provided datasets:

Civil Aviation Administration of China (CAAC) statistics: Monthly key operating indicators, annual statistical bulletins on civil aviation industry development, and annual airport operating statistics. Link: https://www.caac.gov.cn/XXGK/XXGK/TJSJ/index_1215.html

CAAC News airline and major airport punctuality statistics: Average flight punctuality rates for domestic passenger airlines, average on-time departure rates for major airports, and explanations of the relevant categories. Search the site for “Statistical results for punctuality rates of domestic passenger airlines and major airports” (in Chinese). Link: http://www.caacnews.com.cn/

China Meteorological Data Service Centre: Surface weather observations, historical monthly values, current weather conditions and forecast data. Some data may require registration; availability is subject to the information on the website. Link: https://data.cma.cn/site/index.html

China Civil Airports Association airport operations data: An additional source for finding industry bulletins and airport operating information. Link: https://chinaairports.org.cn/dl/lpUJnj6vdi.html

Official websites of listed airlines or CNINFO: Use these only for supplementary public airline announcements and background material, not as core data for this challenge. Link: http://www.cninfo.com.cn/new/index.jsp

The organizers will provide a set of monthly airport operating baselines, annual airport size tables and simulated events, together with a standard output template for the AI assistant, test questions, a human feedback form and scoring requirements.